Swayam Gupta

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SUMMARY

Backend engineer with Python, Node.js, and SQL/NoSQL experience across production systems — REST API design, data pipelines, and service reliability. Built a self-hosted middleware ecosystem with circuit breakers, rate limiting, and real-time observability tooling, plus Python/FastAPI backends powering multi-agent AI systems.

EDUCATION

Jaypee Institute of Information Technology **Noida, Uttar Pradesh**
Bachelor of Technology, Computer Science and Engineering | CGPA: 7.84/10 *July 2027*

PROJECTS

Mirai - Real-Time Multi-Agent AI Collaboration Platform (Python + GenAI) - https://mirai-pi.vercel.app | Open Source
Full-Stack / AI | Python, FastAPI, Next.js, WebSockets

- Architected and deployed a decoupled full-stack system (Next.js on Vercel, FastAPI on Render) with OAuth authentication and manual rate limiting, maintaining stable multi-service performance under concurrent user load.
- Built a real-time collaborative AI chat system using WebSockets with shared context, live message synchronization, and streaming responses, ensuring sub-second latency communication and consistent message edit and delete behavior during concurrent usage.
- Designed and built the backend API for a CrewAI-based multi-agent platform using FastAPI, coordinating specialized agents for reasoning, coding, and writing; onboarded 50+ users in week one.
- Open-sourced the project on GitHub, maintaining active issues and documentation to onboard external contributors.

Medical/Academic PDF RAG System (Python / FastAPI) - https://github.com/SwayamGupta12345/Medi_back
Backend / AI Infra | FastAPI, Pinecone, Gemini, LangChain, CrewAI, PyMuPDF

- Built a document QA system over medical PDFs, starting with a custom ingestion pipeline that strips watermark and boilerplate text using font size, rotation, and opacity heuristics, automatic title extraction from PDF metadata and font-based heading detection.
- Implemented Reciprocal Rank Fusion for retrieval, combining dense vector search over 3072-dimensional Gemini embeddings with keyword-overlap ranking instead of relying on cosine similarity alone.
- Added batched, rate-limited embedding calls with exponential backoff after hitting live Gemini API quota limits during bulk PDF uploads, along with embedding-dimension validation at both ingest and query time.
- Built a retrieval evaluation harness measuring precision@k, recall@k, MRR, and NDCG against a 40-query labeled set, achieving 95% MRR and 100% recall@5, to validate hybrid retrieval quality instead of relying on spot checks.

Jaypee Learning Hub - Academic Resource Management Platform (Full-Stack) - https://jaypeelearninghub.great-site.net
Web Development

- Built and scaled an academic platform to 1000+ active users within 4 months, managing 1200+ resources.
- Architected the entire data model and backend logic independently, later refactoring database schemas and optimizing SQL queries to improve load times by 30%, data consistency, and overall system responsiveness.
- Structured and designed pages using semantic HTML and responsive CSS supporting screen widths down to 240px, from rough wireframe sketches, built entirely in vanilla JavaScript, CSS, and PHP with no frameworks.

WORK EXPERIENCE

Style My Space (AI Research & Product Intern) - **Remote** *Jan 2026- March 2026*

- Identified the team's need for a repeatable way to evaluate outputs across generative models, then scoped, built, and shipped a full-stack evaluation platform (Next.js, MongoDB, Cloudflare R2) that the team adopted directly into its workflow.
- Defined evaluation criteria and ran structured tests across 50+ prompt variations and multiple generative models, turning results into a data-driven model selection framework covering prompt adherence, output clarity, structural correctness, and consistency.
- Owned the product end to end as sole engineer, from requirements through interface design, building a comparison dashboard that handled 1,000+ generated assets with pagination and lazy loading so the team could review results without lag.
- Set naming and versioning conventions for asset handling across the platform, cutting duplication and retrieval friction, and built prompting interfaces that shortened the team's testing turnaround on repeated iterations.

SKILLS

Tools and Platforms: Git, GitHub, Docker, DockerHub, Linux (CLI, Bash), Pinecone, CrewAI
Data & Analytics: SQL (MySQL), NumPy, Pandas, Scikit-learn, Data Preprocessing, Matplotlib, Model Training, RAG, LangChain
Programming Languages: Python, C/C++, JavaScript, PHP
Web & Backend Technologies: HTML, CSS, DOM, Node.js, React, Express.js, FastAPI, REST APIs, MongoDB, Next.js
Soft Skills: Agile Project Management, Cross-functional Collaboration, Technical Leadership, Public Speaking

ACHIEVEMENTS

- Solved 670+ problems on LeetCode (Swayam_Gupta) with a contest rating of 1,721; Codeforces (swayam12) rating 1,324 (Pupil).
- AWS Cloud Practitioner coursework completed Certificate .
- McKinsey Forward Program - strategic thinking, agile problem-solving, leadership, and communication. Certificate Badge.
- Technical Head, AI/ML Club - Organized a club-level hackathon and mentored peers on ML and system design.
- Completed Krish Naik's Data Science, ML, DL & NLP Bootcamp on Udemy. Certificate